7 Extended Data — COHA Timeseries for Type Token Ratio and Zipf exponent

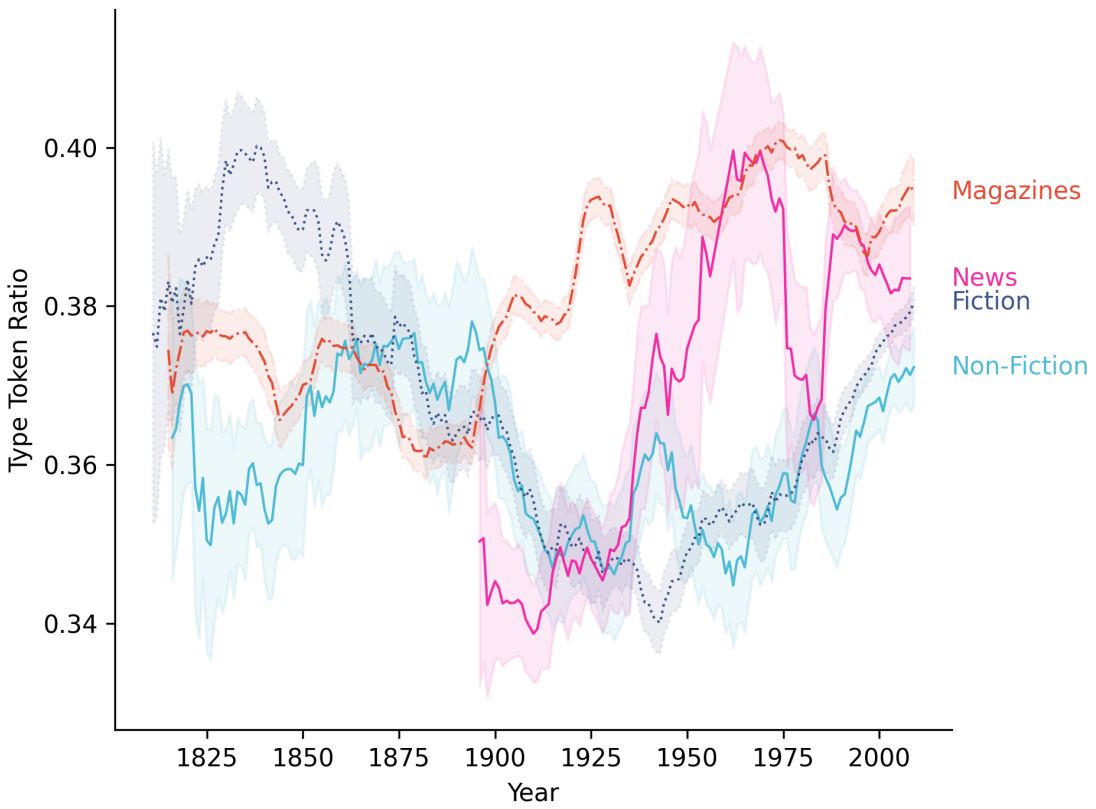

Figure 2: Historical timeseries of type token ratio in the Corpus of Historical American English. Type token ratio was calculated for text samples from COHA truncated with $N = 2000$ words. For each media category and year, a moving average of all valid samples with ± 5 years was calculated. The shaded region shows a 95% confidence interval for this average.

8 Extended Data — Corpora Boxplot Distributions for Word Entropy, Type Token Ratio and Zipf exponent

9 Supplementary - Timeseries Breakpoint Analysis

As discussed in Methods, we carried out a piecewise-regression analysis on the median annual values for each of the lexical measures and media categories (Figure 6). With the type token ration for the News media category, the breakpoint was found close to the edge of the data. If we restrict the position to avoid being close to the edge then the breakpoint is estimated in a similar location as to the Word Entropy and Zipf exponent. The short-form media shows signs of a rise in lexical diversity before long-form media, consistent with the model in the main paper.

We ran the same analysis with the media categories collated to give an average mean each year (Figure 7). Notably, the confidence interval for the breakpoint includes the year 1900.

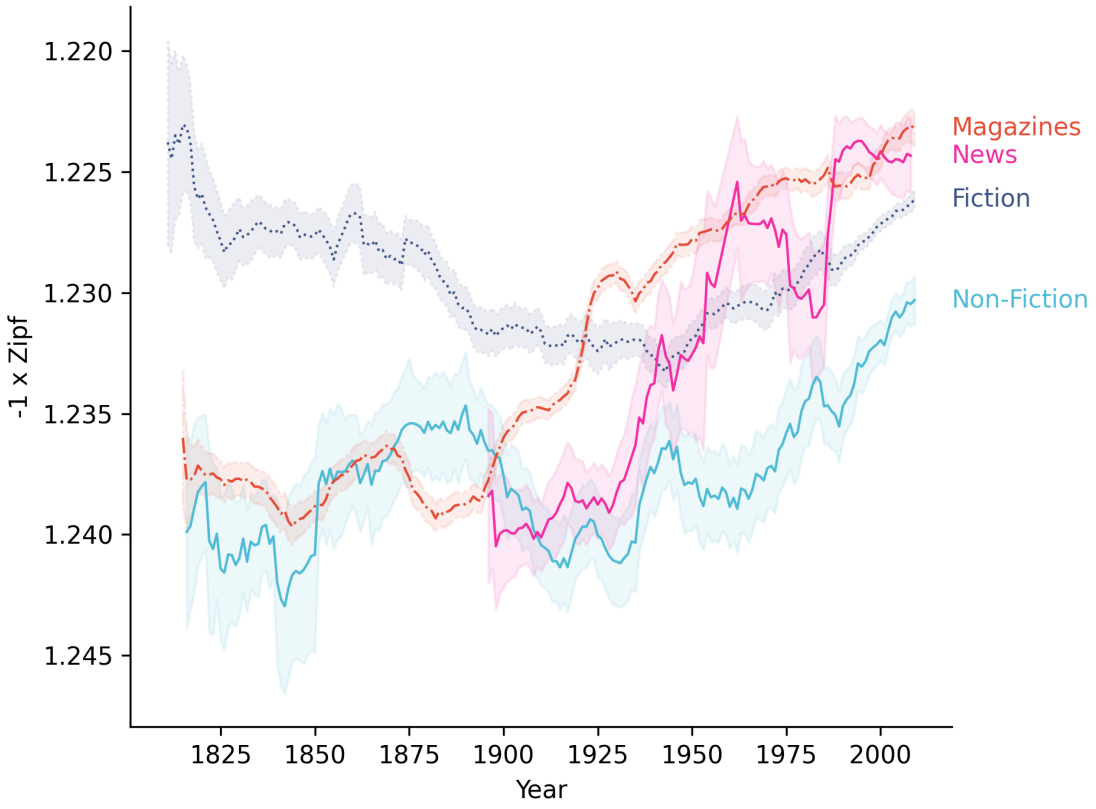

Figure 3: Historical timeseries of Zipf exponent in text samples in written media categories in American English. The timeseries was calculated in the same way as in the previous figure.

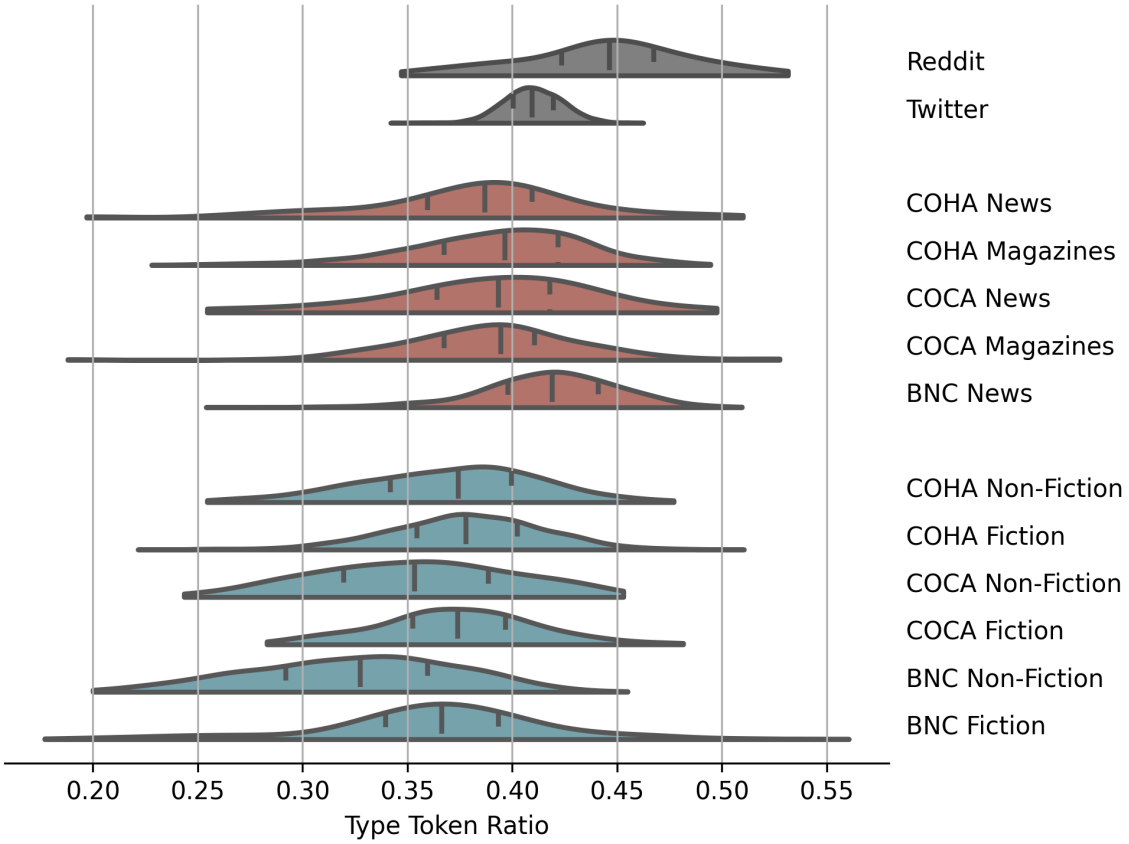

Figure 4: Distribution snapshots of type token ratio across different text corpora for text samples with $N = 2000$ words. COHA samples are from the year 2000 onwards only. Social media text samples were collated from status updates.